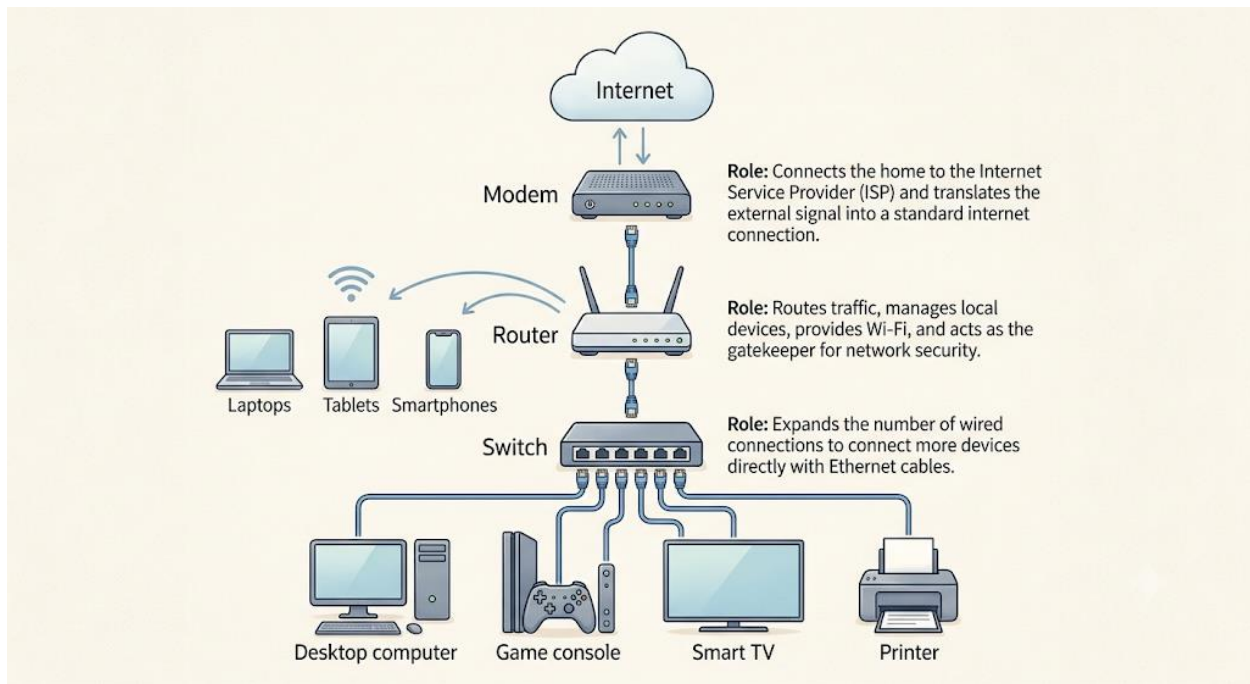


ASSIGNMENT : 12 – ROUTER & MODEM & SWITCHES

GUIDED BY : VIPUL MISTRI

Q.1 Draw a simple network diagram showing how a router, modem, and switch connect together in a typical home setup. Label each device and explain the role of each in one line on your diagram.

ANS :



Q.2 Log in to your home WiFi router's admin panel (usually at 192.168.0.1 or 192.168.1.1) and find the section that shows connected devices. List the names or MAC addresses of at least 3 devices currently connected to your network. Hint If you don't have access to a physical router, search online for a router admin demo page and use that for this task.

ANS : here is a breakdown of what a router's **Connected Devices / DHCP Client List** looks like using a standard TP-Link/Netgear router interface:

Example Router Connected Devices Table

Device Name	IP Address	MAC Address	Connection Type
DESKTOP-MAIN-PC	192.168.1.100	00:1A:2B:3C:4D:5E	Wired (Ethernet)

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Device Name	IP Address	MAC Address	Connection Type
Galaxy-S23-Ultra	192.168.1.102	A4:C3:F0:11:22:33	Wireless (5 GHz Wi-Fi)
HP-LaserJet-Printer	192.168.1.105	70:B3:D5:12:34:56	Wireless (2.4 GHz Wi-Fi)

How to Check This on Your Own Router:

1. Open your web browser and go to your **Default Gateway** IP (typically 192.168.1.1 or 192.168.0.1).
2. Log in with your router's admin credentials.
3. Look for a tab named **DHCP Client List**, **Connected Devices**, **Network Map**, or **Attached Devices**.
4. You will see every active phone, PC, printer, and smart device currently assigned an IP on your local network.

Q.3 Create a table comparing routers, modems, and switches. Include at least 4 points for each: main function, where it is used (home/office), whether it assigns IP addresses, and if it supports WiFi.

ANS : Here is the comparison table for routers, modems, and switches across the requested key parameters:

Feature / Criteria	Modem	Router	Switch
Main Function	Converts (modulates/demodulates) ISP signals (fiber, coax, or DSL) into digital Ethernet data.	Directs data traffic between different networks (your local network and the internet)	Connects multiple end-devices on the same local network (LAN) using MAC addresses to

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Feature / Criteria	Modem	Router	Switch
		using IP addresses.	forward data frames.
Where Used	Found in both home and office environments at the internet entry point.	Found in both home and office networks to manage traffic and share the internet.	Primarily used in offices and larger environments (or home labs) that require many wired ports.
Assigns IP Addresses?	No. It receives a single public IP address from your ISP and passes it along.	Yes. Runs a DHCP server to assign local private IP addresses to devices on your LAN.	No. (Standard Layer 2 switches do not assign IP addresses; they only forward traffic).
Supports Wi-Fi?	No. (Standalone modems only provide wired output; ISP combo units combine this with a router).	Yes. Most consumer routers feature built-in Wi-Fi access points.	No. Switches only provide physical RJ45 Ethernet ports for wired connections.

Q.4 Imagine you are setting up a small gaming café. Choose and specify which devices (router, modem, switch) you would need, how many of each, and briefly explain the reason for each choice. Hint: Consider the number of computers, internet connection, and need for wired vs wireless access.

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ANS : Here is the network equipment breakdown for setting up a small gaming café (assuming 15–20 high-performance gaming PCs plus Wi-Fi for staff and customers):

Required Devices & Quantities

Device	Quantity	Specified Model / Type
Modem	1	Standalone Fiber ONT (Optical Network Terminal)
Router	1	Dedicated Gigabit Business Router / Firewall (e.g., Ubiquiti UniFi Security Gateway or Cisco RV Series)
Switch	1 or 2	24-Port Gigabit Managed Switch (e.g., TP-Link TL-SG1024 or Cisco Business 250)
Wi-Fi Access Point	1	Ceiling-Mounted Wi-Fi 6 Access Point (e.g., Ubiquiti UniFi AP)

Reason for Each Choice

1. Modem (1 Unit):

- **Why:** You need a high-speed fiber internet entry point from your ISP. A standalone ONT converts the incoming optical fiber signal into a clean Ethernet connection for your router.

2. Business Router (1 Unit):

- **Why:** Standard consumer all-in-one home routers crash under heavy multi-device traffic. A dedicated business router handles high packet

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processing rates, manages Quality of Service (QoS) to prioritize low latency (ping) for gaming, and provides basic firewall protection.

3. 24-Port Managed Switch (1–2 Units):

- **Why:** Online gaming requires low latency and zero packet loss, which wireless connections cannot guarantee. Connecting every gaming PC directly to a wired switch via Cat6 Ethernet cabling gives each PC maximum speed and stability. A **managed** switch allows you to set up VLANs to separate gaming traffic from customer Wi-Fi.

4. Dedicated Wi-Fi Access Point (1 Unit):

- **Why:** Instead of relying on Wi-Fi built into a router (which might be placed in a back server rack), a dedicated ceiling-mounted Access Point provides clean wireless internet for customer mobile phones, staff, and POS systems without dragging down the bandwidth of the gaming PCs.